Practice workbook · Bubble sort, one pass at a time

Showing the state of a list after each bubble sort pass, and counting swaps or passes.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-08

1. Question 3 · part a.i · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.i only.

DO NOT WRITE IN THIS AREA

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3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

1. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b.i · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part b.i only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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2. Question 5 · part b.i · 3 marks

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Further practice · Attempt part b.i only.

(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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P 7 2 4 0 6 A 0 1 1 1 6

2. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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3. Question 3 · part a.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a.ii only.

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

3. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 3 · part a.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a.iii only.

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The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

4. Question 3 · part a.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a.iii only.

(b) The youth club stores a list of sports equipment.

The list contains over 1000 items, sorted in ascending order.

State and justify the **most** appropriate search algorithm to find an item in the list.

(2)

Search algorithm

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Justification

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4. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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